

Sunoikisis DC Digital Approaches to Cultural Heritage
Session 6

Working with Indigenous Heritage

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Outline

1. Definitions
2. FAIR and CARE Principles
3. Case study #1: Documenting Indigenous knowledge(s) in a digital repository
4. Case study #2: Working with Indigenous Archaeology in Canada
5. Case study #3: Digitising culturally sensitive Sámi heritage: the case of the ancestral drum
6. Granting bodies and indigenous heritage
7. Exercise

Definitions



Indigenous



Cultural Heritage and Knowledge



Digital Technologies

Definitions

Indigenous

- ★ No single universal definition: varies by regions, continents and context
- ★ Organized groups: communities, tribes, clans, nations
- ★ Diverse ties to a region/ territories; predate colonization/ settler states
- ★ Maintain distinct cultures, languages, social, political and intellectual systems
- ★ Here in the Circumpolar Arctic: Sámi communities span Norway, Sweden, Finland and Russia (Kola Peninsula)



Photo: diverse North Sámi baby cradles in the collections of RiddoDuottarMuseat – Sámiid Vuorká-Dávvirat. Photograph by Paula Rauhala, 2025.

Definitions

Cultural Heritage and Knowledge

- ★ Tangible and intangible aspects
- ★ Social, practical, spiritual, intellectual, aesthetic domains
- ★ Traditional and contemporary expressions
- ★ From everyday utensils to esoteric/ culturally sensitive heritage
 - use, production/ acquisition, historical context
- ★ Indigenous knowledge systems
 - relational
 - Indigenous ontologies
 - epistemologies; epistemic plurality
- ★ Ancestral North Sámi drums; non-human beings
 - posthumanism

Photo: the reverse side of a replica of an ancestral Sámi drum, in the collections of RiddoDuottarMuseat – Sámiid Vuorká-Dávvirat. Crafted by Osvald Guttorm.
Photograph by Paula Rauhala, 2025.
© RiddoDuottarMuseat – Sámiid Vuorká-Dávvirat



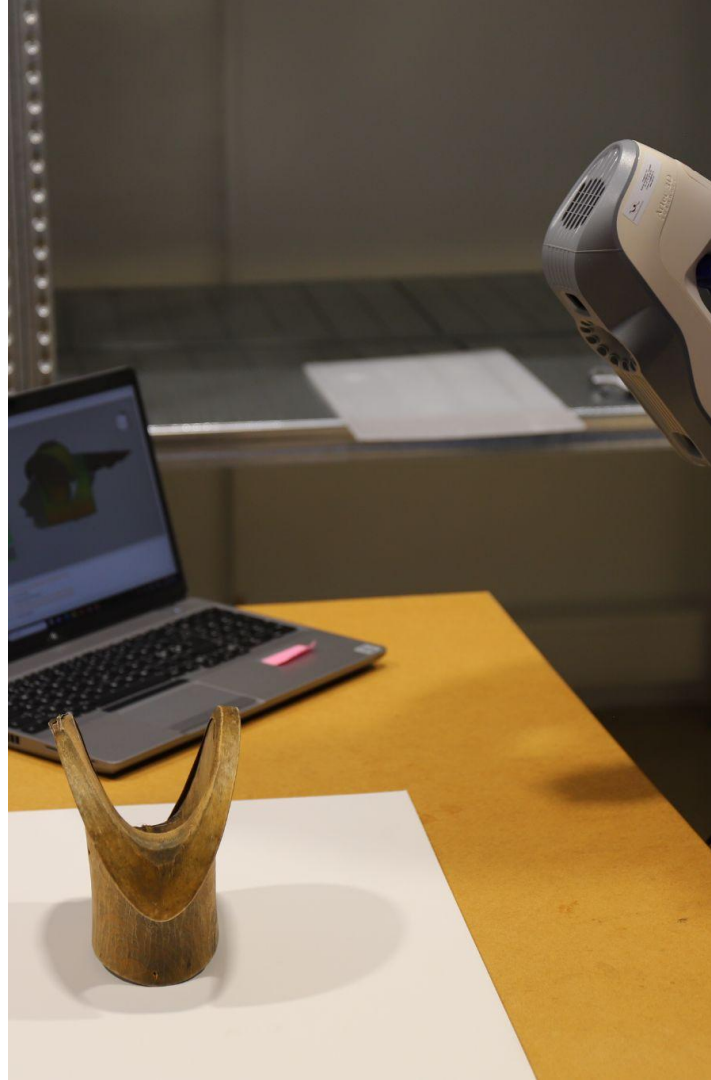
Definitions

Digital Technologies

- ★ 2D and 3D practices
 - 2D: photography, illustrations, digital scans, audiovisual recordings
 - 3D: photogrammetry, laser scanning, structured-light scanning, CT, ToF cameras, LiDAR
 - 3D printing
- ★ GIS and mapping
- ★ Reflectance Transformation Imaging (RTI)
- ★ Use of 3D technologies in Indigenous cultural heritage contexts:
 - Tlingit crest hat: Sitka Tlingit clan, Alaska: in collab. w/ the Smithsonian Institution
 - Nunalleq Digital Museum – Yup'ik heritage online
 - Digitization of an ancestral Sámi drum at RiddoDuottarMuseat–Sámiid Vuorká-Dávvirat

Photo: Jelena Porsanger 3D scanning a wooden piece from a Sámi woman's headdress, Oslo, 2023.

© RiddoDuottarMuseat – Sámiid Vuorká-Dávvirat



Working with Indigenous/Community Knowledge(s), Heritage and Data

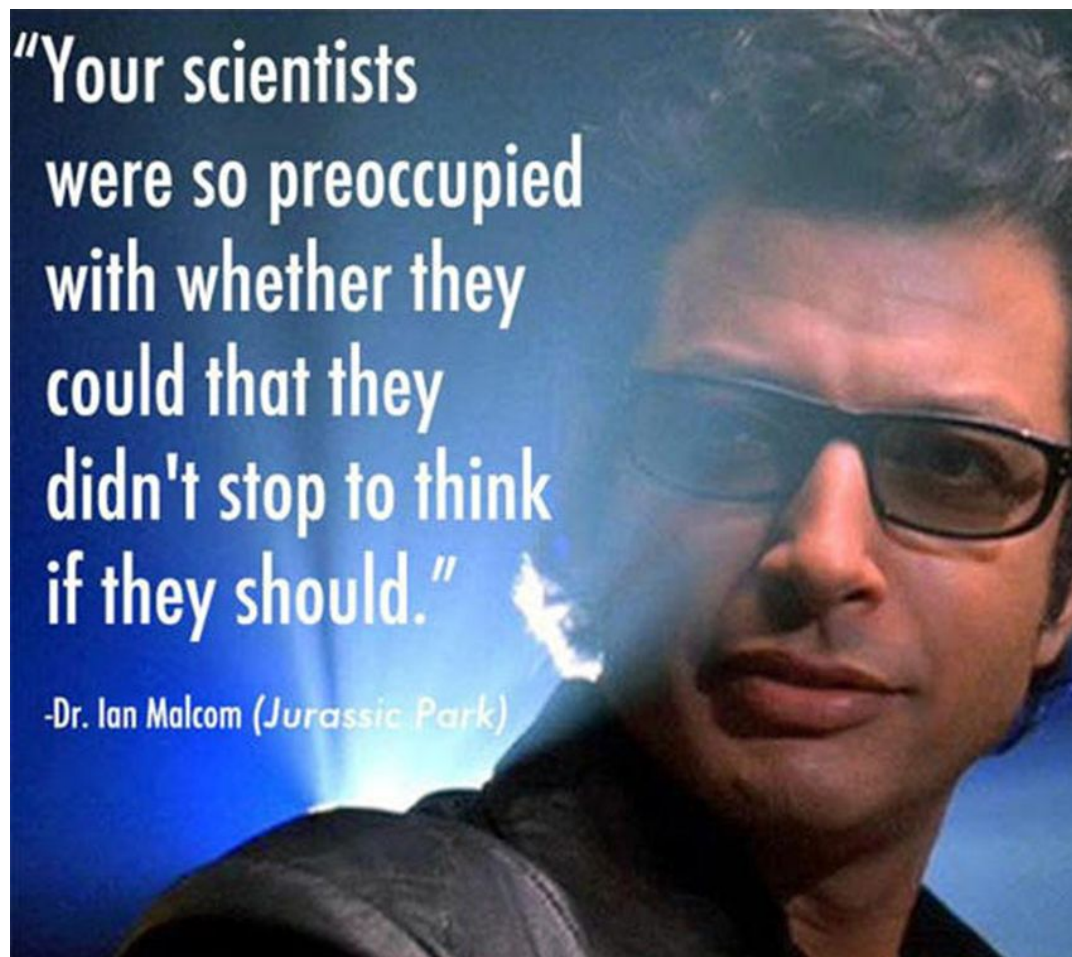
Guiding Questions

- Who are we doing this for?
- Who is involved?
- What are the benefits?
- What is the impact?
- Who is the author/creator?
- How will it be shared?



**"Your scientists
were so preoccupied
with whether they
could that they
didn't stop to think
if they should."**

-Dr. Ian Malcom (*Jurassic Park*)



FAIR + CARE PRINCIPLES



CARE Principles for Indigenous Data Governance (Research Data Alliance International Indigenous Data Sovereignty Interest Group, 2019).

The CARE principles are meant to **redress the power imbalance between communities, data collectors, and digital service providers**. The data and data storage should be designed to:

- Enable indigenous communities to benefit from the data
- Recognise indigenous authorship of the data, their rights and interests in it, and data should be actively stewarded with/by community members
- Facilitate responsible use and **expand capacity in digital literacy and infrastructure within indigenous communities** through reciprocal relationships, generating data in the languages and worldviews of those communities
- Minimise harm, respect the rights and fairly represent those communities
- Take into account possible future uses of the data.

DATA GOVERNANCE AND CARE PRINCIPLES

Practice 'CARE' in data collection	Engage 'CARE' in data stewardship	Implement 'CARE' in data community	Use 'FAIR' with 'CARE' in data applications
Define cultural metadata Record provenance in metadata	Use appropriate governance models Make data 'FAIR'	Indigenous ethics inform access Use tools for transparency, integrity and provenance	Fairness, Accountability, Transparency Assess equity

Carroll, S.R., Herczog, E., Hudson, M. et al. Operationalizing the CARE and FAIR Principles for Indigenous data futures. *Sci Data* 8, 108 (2021). <https://doi.org/10.1038/s41597-021-00892->

Collective Benefit

- Whose heritage?
- Who is making the decisions?
- Who will benefit from this?
- Who will take long ownership?
- How will you enable multiple voices to be reflected in the creation and presentation of 3D models?



Workshop on 3D methods for cultural heritage documentation in collaboration with EMKP and the Centre for Digital Skills in Visual and Material Culture at University of Brighton .

Authority to Control

- Whose consent is required?
- Who is taking the lead?
- Who are you planning to involve/consult with?
- How will you ensure long-term control once the materials have been published/released?
- If you are considering publishing results in open access, how are you communicating the implications of open access to your participants?



Doriano Morales taking photos of an object to create a 3D model as part of his EMKP project, MakerSpace, Digital Humanities Research Hub, University of London.

Responsibility and Ethics

- How are you going to ensure the development of positive relationships with the knowledge holders, grounded in respect, reciprocity and mutual understanding as defined by the communities of origin?

Data lifecycle



The data lifecycle © by the 3D Service Suite,
University of Brighton under CC BY 4.0

Endangered Material Knowledge Programme



“[U]nderstanding of the resources, skills, technologies and social values necessary to create and maintain the material world around us”

(www.emkp.org)

Recording (and celebrating) process rather than product



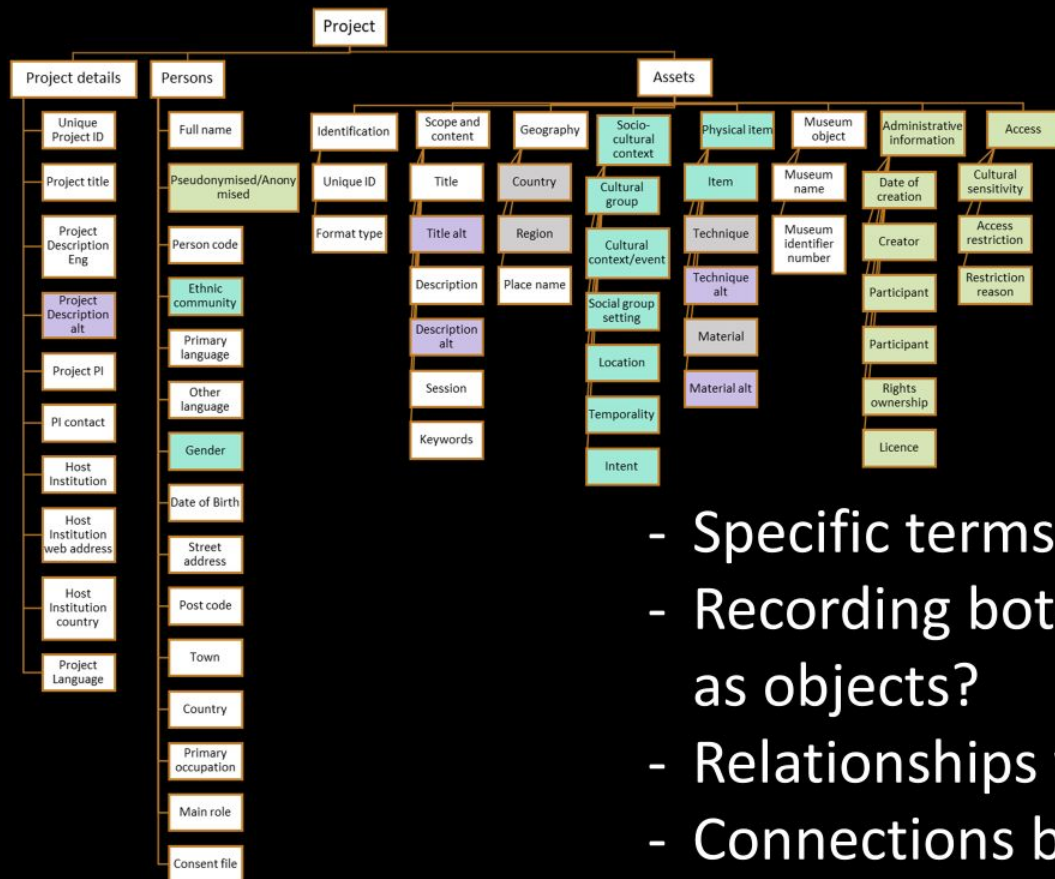
Capturing living knowledge



- Documentation that focuses on movement, gesture and embodied knowledge, and the creator's interaction with the material world.
- Technology offers the possibility to contextualise knowledge.
- Multivocal contexts that require collaborative environments for production and curation.

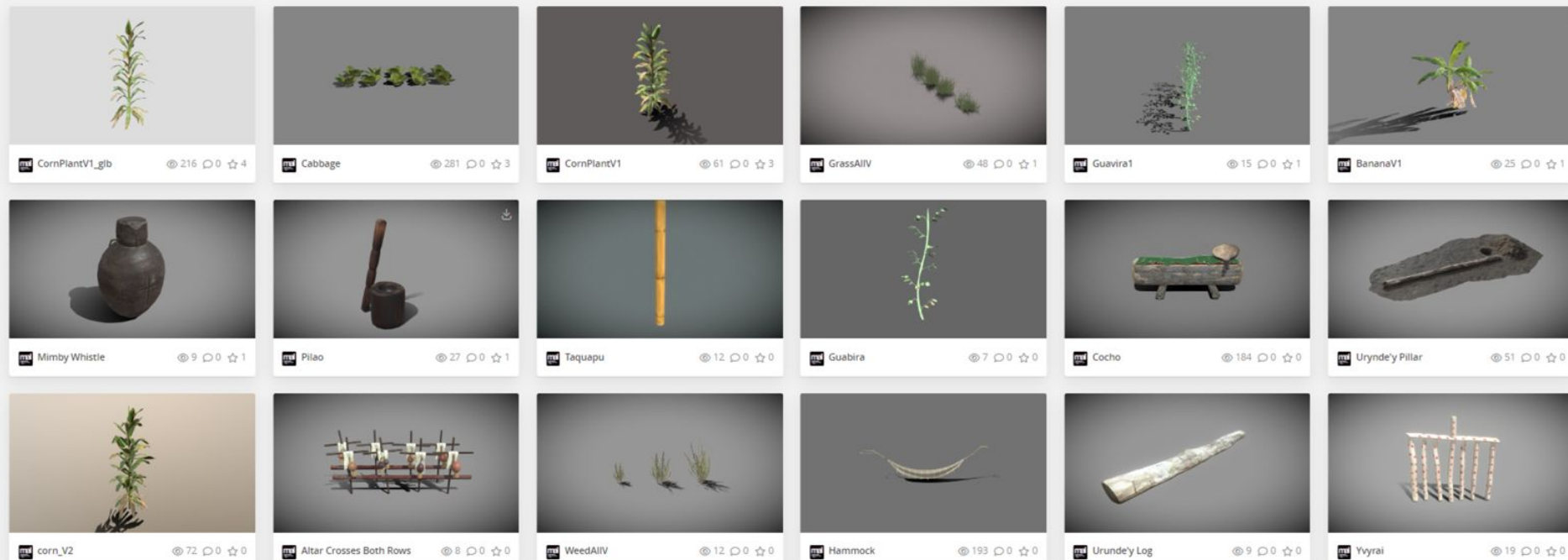
Photo: Maria Luisa Lucas

Representing Diverse Ontologies



- Specific terms and vocabularies
- Recording botanical knowledge – plants as objects?
- Relationships with the environment
- Connections between places/assets

Representing Diverse Ontologies



Raffaella Fryer-Moreira Oga Pysy: Guarani
and Kaiowá Ceremonial Houses in Mato Grosso do Sul, Brazil

Capturing language and knowledge



Notes on Batek pandanus weaving

Alice Rudge

Types of pandanus (hakeʔ)

<i>hakeʔ ciŋ</i>	type of pandanus, requires drying over a fire before weaving, tougher and more water resistant as well as smaller and thinner leaves, so better for <i>cok</i> ('small bag')
<i>hakeʔ hatiŋ</i>	type of pandanus, grows from a tree trunk, goes yellow when dried
<i>hakeʔ kajaŋ</i>	type of pandanus
<i>hakeʔ kayuʔ</i>	type of pandanus
<i>hakeʔ panen</i>	type of pandanus, requires soaking before weaving, usually used by Malays and is commonly planted
<i>hakeʔ pɔrɔn</i>	type of pandanus, fades to <i>blʔaŋ</i> ('blueish') colour, and needs to be stamped on before it can be used, best for making a <i>cok</i> ('small bag') because it is tougher.
<i>hakeʔ ʔular</i>	type of pandanus, cannot be used for weaving because it is <i>lawac</i> ('set of taboos that risk angering thunder being', see Endicott (1979) and Rudge (2019)).



Alice Rudge, list of terms related to Pandanus production for the “project “Material Culture of Batek Hunter-Gatherers in Pahang State, Malaysia”.

Changes in community perspective towards the sharing of data



Maria Luisa Lucas - The Bora design system: documenting the knowledge networks of designs made on bodies and objects in an Amerindian ritual

Ecological Impact



- Ecological knowledge is a fundamental part of material knowledge.
- Measure the impact of our activities
- Consider alternative ways in which the ecological impact can be reduced
- Assess the potential direct and indirect consequences on the environment against the long-term good generated by our projects.



Foggaras, water management systems, Algeria.
Juan Ignacio Robles, EMKP grantee 2023.

Working with Indigenous Archaeology in Canada

Working with Indigenous Archaeology in Canada



Nicole Iu

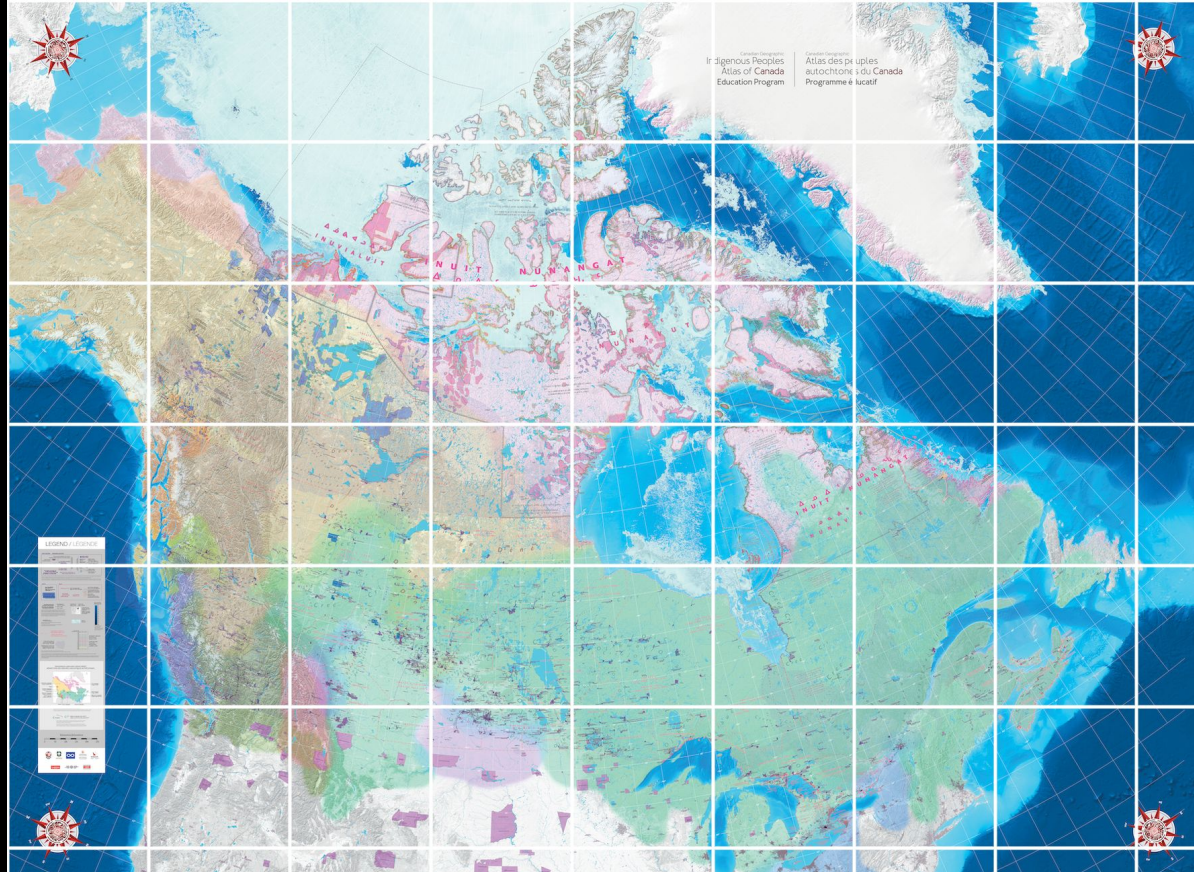
Introduction

- Reconciliation Mandate: “Building a renewed relationship with Indigenous Peoples based on the recognition of rights, respect and partnership”
- Indigenous Stewardship: “Parks Canada workers and Indigenous partners will make the plans together. The plans will be unique to each place and shaped by the local context centering Indigenous systems of knowledge, protocols, and approaches. Parks Canada will work with Indigenous knowledge holders and communities to find culturally appropriate ways to use Indigenous knowledge and knowledge systems and to foster cultural continuity... Humans have a duty and a responsibility to live in balance with all other living things and to respect and care for the lands, water, and ice.”

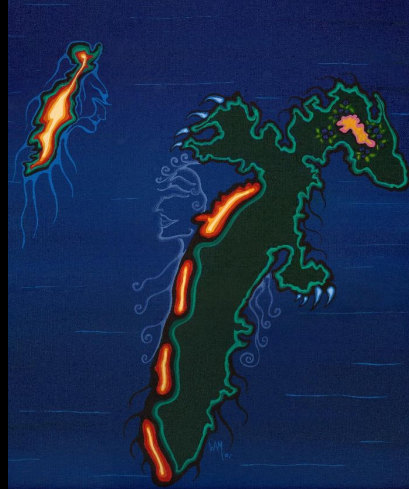


Source: Government of Canada

Indigenous groups in Canada



Case Study: Beausoleil Island



- Georgian Bay Islands National Park
- Traditional territory of the Anishinaabeg
- At the “crossroads” of many different groups for trade, interaction
- Unique pottery typology

Source: Parks Canada; art provided to Parks Canada by the Montague family

Protocols when working with Indigenous Peoples

- Tobacco offerings
- Groundbreaking and closing ceremonies
- Water Ceremonies and Talking Circles
- Smudging



Final Thoughts

- Greater awareness regarding the use of the term 'Romanisation' in archaeology
- Not oversimplifying or over-generalising indigenous groups, in Canada, today, nor in the ancient world
- Diversify perspectives and accounts, avoid misinterpretation based solely on a Western models
- Indigenous archaeology is often connected and meaningful to many people today, and not simply an object of study

Digitising Culturally Sensitive Sámi Heritage: The Case of the Ancestral Drum

Paula Rauhala

RiddoDuottarMuseat—Sámiid Vuorká-Dávvirat



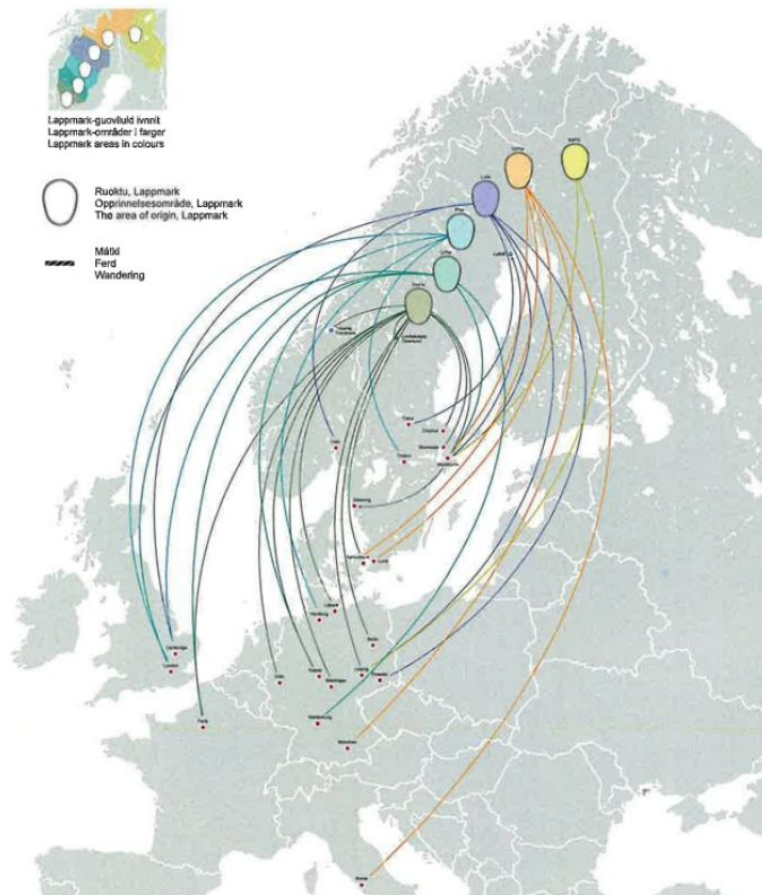
Background on Sámi Drums

- ★ Sámi drums were expropriated from the communities during 1600s, 1700s and 1800s
- ★ Sámi drums embody spiritual and practical traditions, carrying both tangible and intangible heritage: drums served as companions for community healers and ritual specialists
- ★ Non-human beings vs. Cartesian dualism
- ★ 72 ancestral Sámi drums have been discovered globally
- ★ Many Sámi drums are currently held in museums across Europe and the UK

Photo: the estimated paths of some expropriated Sámi drums. Sámiid Vuorká-Dávvirat Catalogue, 2022.

© RiddoDuottarMuseat – Sámiid Vuorká-Dávvirat

Gos rumbbut leat vižžon?
Hvor ble trommene tatt fra?
The estimated paths of some expropriated drums



Creating a 3D Digital Representation of Paul-Ánde's Goavddis

- ★ Goavddis = North Sámi term for a bowl-shaped drum crafted from a large and naturally grown, hollowed-out pine burl
- ★ Paul-Ánde
 - Sámi reindeer herder and healer
 - born ~1600
 - 1691: Paul-Ánde arrested for suspected witchcraft
 - 1692: sentenced to death, but killed in custody before execution
- ★ Paul-Ánde's goavddis
 - his goavddis presumably made before 1650s
 - goavddis confiscated and sent to Copenhagen in 1692-1694 → Royal Danish Cabinet of Curiosities
 - 1800s: goavddis included in the National Museum of Denmark collection
 - since 1970s: goavddis on loan to RiddoDuottarMuseat – Sámiid Vuorká-Dávvirat
 - 2022: goavddis repatriated to RiddoDuottarMuseat – Sámiid Vuorká-Dávvirat

Photo: 3D scan of Paul-Ánde's goavddis, RiddoDuottarMuseat – Sámiid Vuorká-Dávvirat, 2022.

© RiddoDuottarMuseat – Sámiid Vuorká-Dávvirat



Creating a 3D Digital Representation of Paul-Ánde's Goavddis

- ★ 2020: digitization of the goavddis
 - structured-light 3D scanners
- ★ Process and guidelines
 - institutional guidelines established: how the drum is (1) approached (2) documented (3) managed (4) shared
 - guided by CARE principles
 - acknowledging historical practices and drum's cultural integrity
 - files stored according to cultural status and local Sámi names
- ★ Publishing and access
 - other ancestral drum 2D photos previously published by other museums on DigitaltMuseum.no and museum websites: protected by copyright/ IPR
 - 3D models of ancestral Sámi drums digitized by Sámiid Vuorká-Dávvirat (the Sámi museum) are not openly accessible to prevent cultural commodification and misuse
- ★ Challenges and considerations
 - esoteric elements (e.g.: drum symbols) are often appropriated by businesses
 - emerging risks with 3D printing and new technologies
 - safeguarding and ethical use



Photo: Paul-Ánde's goavddis at RiddoDuottarMuseat – Sámiid Vuorká-Dávvirat, RUOKTOT exhibition, 2022.

© RiddoDuottarMuseat – Sámiid Vuorká-Dávvirat

Conclusion

- ★ Use of 3D technologies for esoteric Sámi cultural heritage is still new
 - need for careful consideration
- ★ Modern digital era brings esoteric Sámi heritage to new digital, social, and political domains
 - need for renewed attention
- ★ Culturally sensitive Indigenous heritage is not the easiest material to approach
- ★ Digitization of esoteric Indigenous heritage: context specific
 - consultation with source communities or relevant parties is recommended
 - using culturally appropriate terminology is recommended
- ★ Digitization of culturally sensitive Indigenous heritage: a holistic process, not just a technical application

Photo: a replica of an ancestral Sámi drum, crafted by Osvald Guttorm. RUOKTOT exhibition. Photograph by Paula Rauhala, 2022. © RiddoDuottarMuseat – Sámiid Vuorká-Dávvirat





Giitu!

Thank you!



Granting bodies and
indigenous heritage

Who we are

Grantmakers for Cultural Heritage Preservation (G-CHeP) is an informal network of funding organisations with a global remit to support projects which protect cultural heritage. Established in New York in 2017, our network fosters information sharing, collaboration, and for impactful grantmaking in cultural preservation.

Objectives:

- Prevent duplication of efforts and projects
- Coordinate emergency heritage protection initiatives
- Address operational challenges and explore grantmaking opportunities
- Evaluate and understand the impact of cultural heritage grantmaking
- Gather data to comprehend funding trends, needs, and demands in cultural preservation.





Academic, Learning



Archaeology



Built Heritage



Community Heritage



Conservation



Culture and Memories



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Last updated 24/07/2025

United Kingdom – Key Grant-Givers for Cultural Heritage

International Philanthropic Foundations



Cultural Emergency
Response (CER)



International & Global Grant-giving organisations



Challenges

- Choose wisely
- Systemic and Procedural Barriers: Complex application processes that exclude applications in specific languages or smaller organisations.
- Cultural Misalignment: Western project-based metrics which fail to capture the holistic and intergenerational value of indigenous "living heritage".
- Rigid Funding Structures: Short-term, project-based grants with restricted flexibility, rigid reporting structures or rigid budget categories.
- Specific requirements: not aligned with community needs (e.g. open access).

Exercise - Review of existing projects

Considering the issues and challenges discussed in this session, select and assess some projects that use digital technologies to record and make available Indigenous heritage and knowledge(s) online.

1. Select two or three examples of projects working with Indigenous heritage and knowledge that are applying digital technologies and are available online.
2. Assess them from the different perspectives and approaches discussed today, considering the FAIR and CARE principles but also access, representation, community involvement, funding etc.
3. Based on your analysis, write a small review about them also considering the different stakeholder perspectives e.g. the developers, the community, and the global user.
4. Make some notes on the different ways or levels at which projects attempt to or succeed in addressing FAIR and CARE concerns, and bring them along for discussion with your next class.